Inverter

Single phase

Inverters

Types of inverters:

- Current Source Inverters
- Voltage Source Inverter

• Single-Phase Inverters

- Half Bridge
- Full Bridge

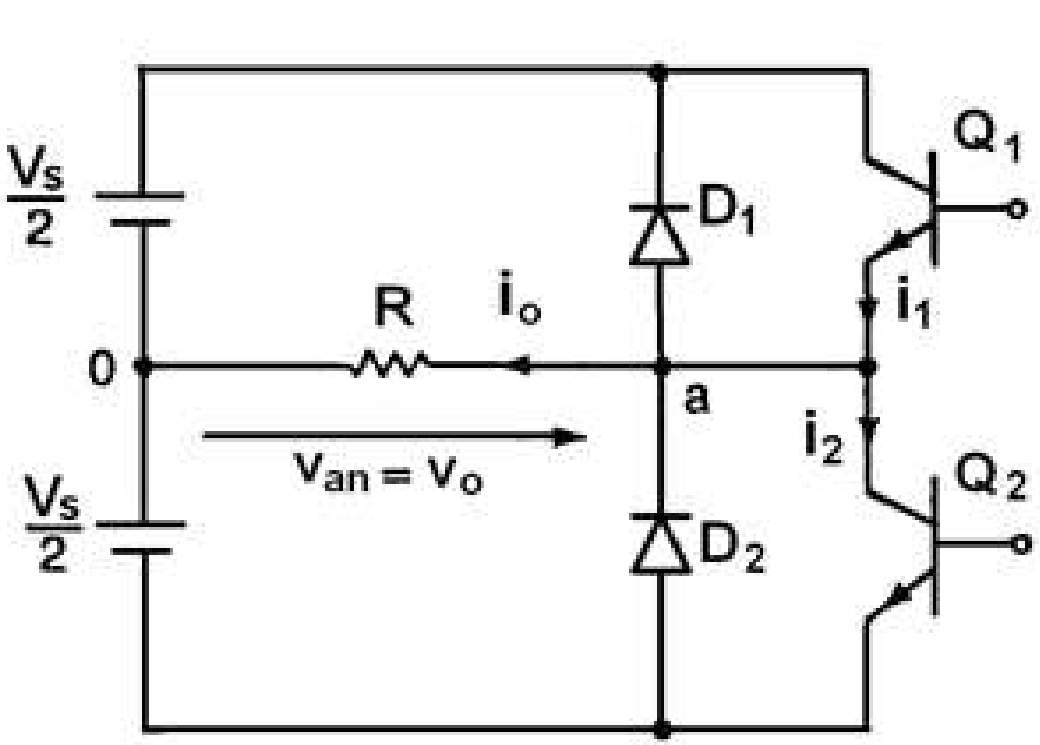
• Three-Phase Inverters

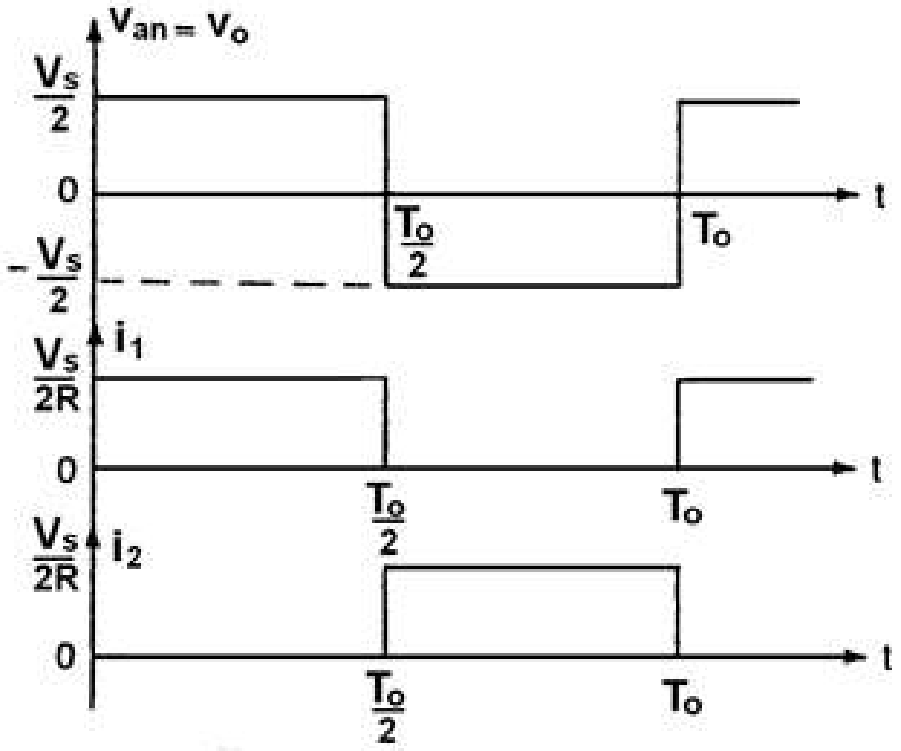
- Half Bridge
- Full Bridge

The output of the inverter

- Square Wave
- Quasi-Square Wave
- Sine-Wave

Single-Phase Half-Bridge Resistive Load Inverter





The rms value of the output voltage is given by:

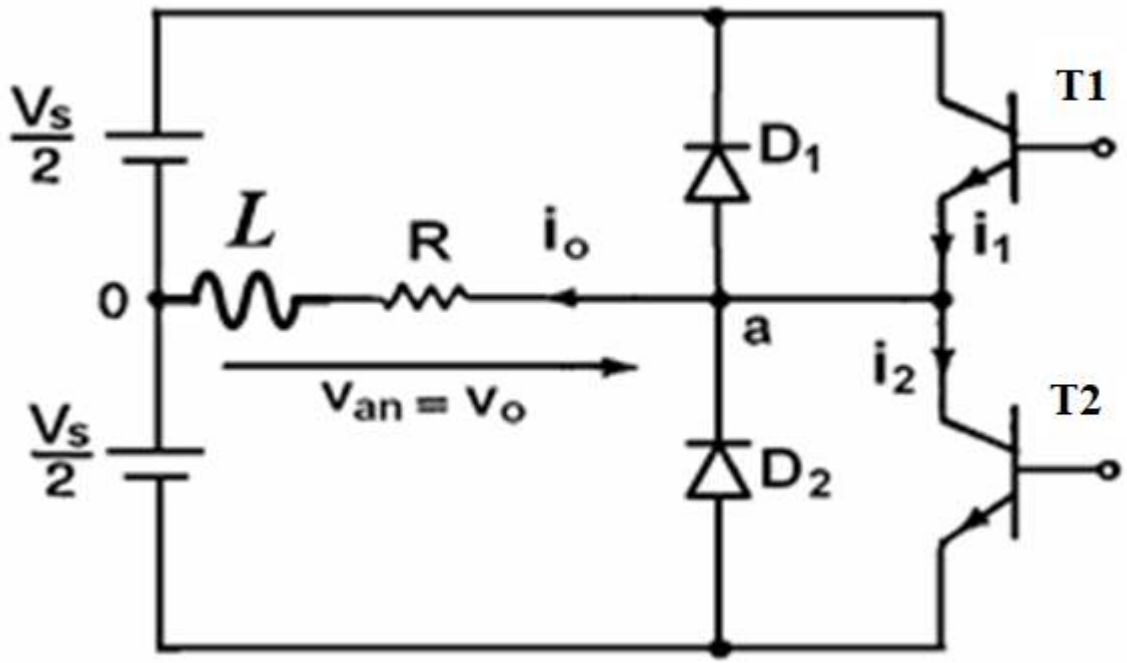
$$V_{rms} = \sqrt{\frac{2}{T}\int_0^{\frac{T}{2}} \left(\frac{V_S}{2}\right)^2 \, dt} = \frac{V_S}{2}$$

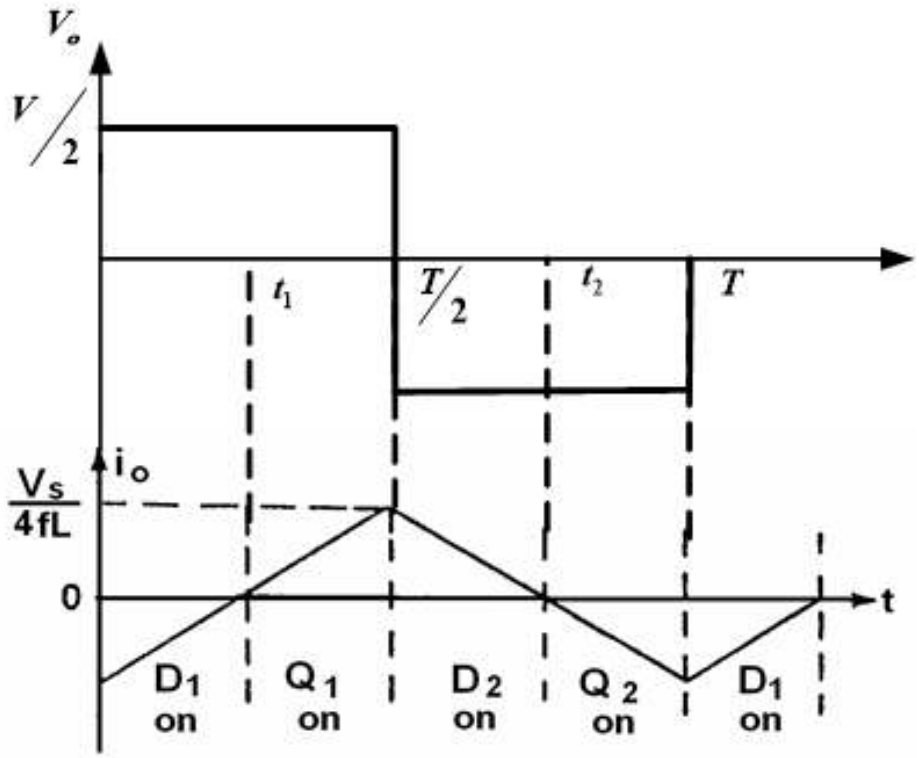
The load current is given by:

$$i_o = \frac{V_o}{R} = \frac{V_S}{2\,R} \qquad 0 < t < \frac{T}{2}$$

$$i_o = \frac{-V_o}{R} = \frac{-V_S}{2\,R} \qquad \frac{T}{2} < t < T$$

Single-Phase half-bridge R L Load Inverter

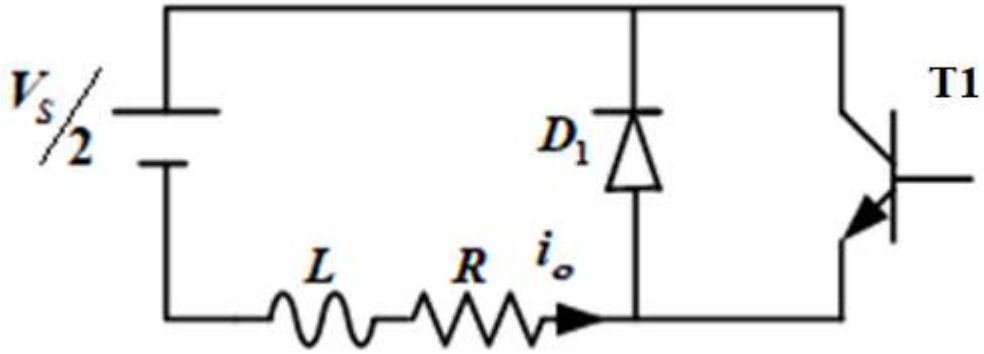




The operation of a single-phase half-bridge inverter with RL load can be divided into four periods:

1- $0 < t < t_1$

At t=0, the control signal is removed from T_2 and a control signal is applied to T_1. At this moment the current is negative maximum. The current can't change the direction directly to positive value due to the inductive load. In this case, the current will flow from the load through D_1 to the source and T_1 stay disconnected in spite of existence the control signal on it due to reverse biased. At $t = t_1$ the current become zero and T_1 start to be forward biased.

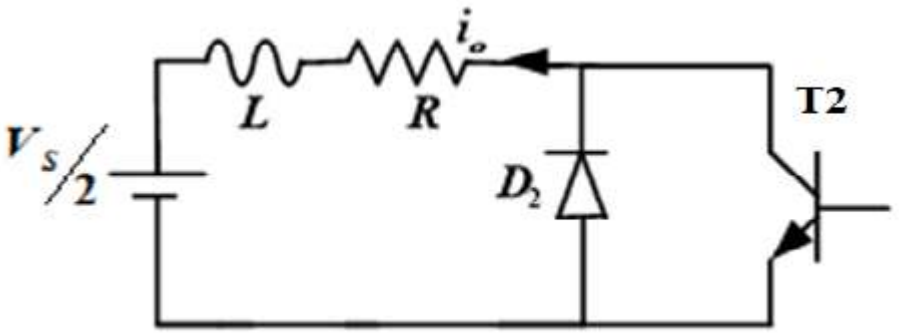


2- $t_1 < t < T/2$

At $t = t_1$, the current change its direction to be positive and T_1 start to conduct. The positive current is increased until reach its positive maximum value at $t = T/2$. At $t = T/2$ the control signal is removed from T_1 and applied to T_2.

3- $T/2 < t < t_2$

At this moment the current is positive maximum. The current can't change the direction directly to negative value due to the inductive load. In this case, the current will flow from the load through D_2 to the source and T_2 stay disconnected in spite of existence the control signal on it due to reverse biased. At $t = t_2$ the current become zero and T_2 start to be forward biased. At this period the voltage become negative and the current positive.



4- $t_2 < t < T$

At $t = t_2$, the current change its direction to be negative and T_2 start to conduct. The negative current is increased until reach its negative maximum value at $t = T$. At $t = T$ the control signal is removed from T_2 and applied to T_1.

The rms value for the output voltage can be found as:

$$V_{rms} = \sqrt{\frac{2}{T}\int_0^{\frac{T}{2}} \left(\frac{V_S}{2}\right)^2 dt} = \frac{V_S}{2}$$

The output current can be found as:

$0 < t < T/2$

$$\frac{Vs}{2} = R\, i_o(t) + L\, \frac{di_o(t)}{dt}$$

$$i_o(t) = \frac{Vs}{2\,R}\left[1 - e^{-\frac{R}{L}t}\right] + I_o e^{-\frac{R}{L}t}$$

At $t = T/2, i_o\,(t) = I_o$

$$I_o = i_o\left(\frac{T}{2}\right) = \frac{V_S}{2R}\left[1 - e^{-\frac{R}{2L}T}\right] - I_o e^{-\frac{R}{2L}T} = \frac{V_S}{2R}\frac{\left[1 - e^{-\frac{R}{2L}T}\right]}{\left[1 + e^{-\frac{R}{2L}T}\right]}$$

Substitute the value of I_o in $i_o(t)$ equation

$$i_o(t) = \frac{V_S}{2R}\left[1 - e^{-\frac{R}{L}t}\right] - \frac{V_S}{2R}\frac{\left[1 - e^{-\frac{R}{2L}T}\right]}{\left[1 + e^{-\frac{R}{2L}T}\right]}e^{-\frac{R}{L}t} = \frac{V_S}{2R}\left[1 - \frac{2\,e^{-\frac{R}{L}t}}{1 + e^{-\frac{R}{2L}T}}\right]$$

$$T/2 < t < T$$

$$t' = t - \frac{T}{2}$$

$$-\frac{V_S}{2} = R\, i_o\,(t') + L\,\frac{di_0(t')}{dt}$$

$$i_o(t') = -\frac{Vs}{2\,R}\left[1 - e^{-\frac{R}{L}t'}\right] + I_o e^{-\frac{R}{L}t'}$$

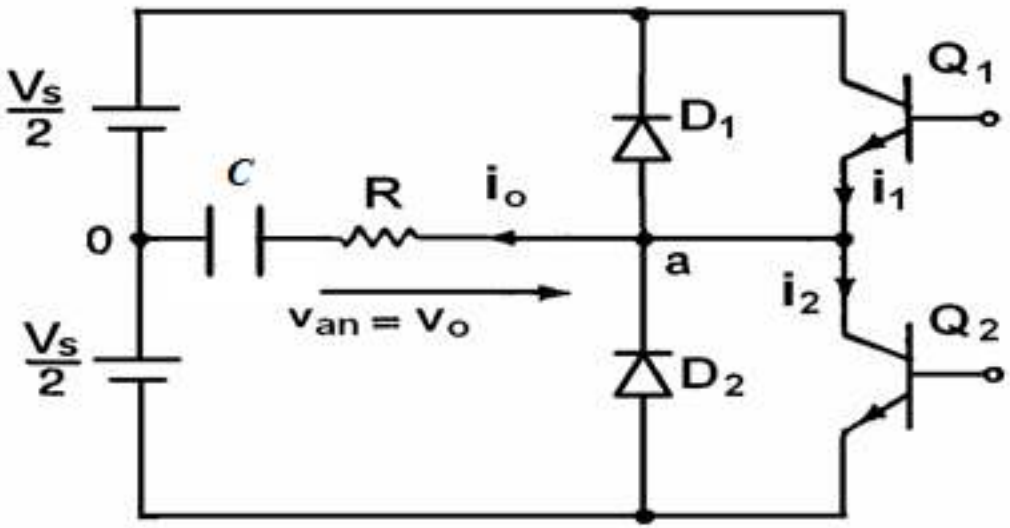
At $t = T/2, i_o\,(t) = I_o$

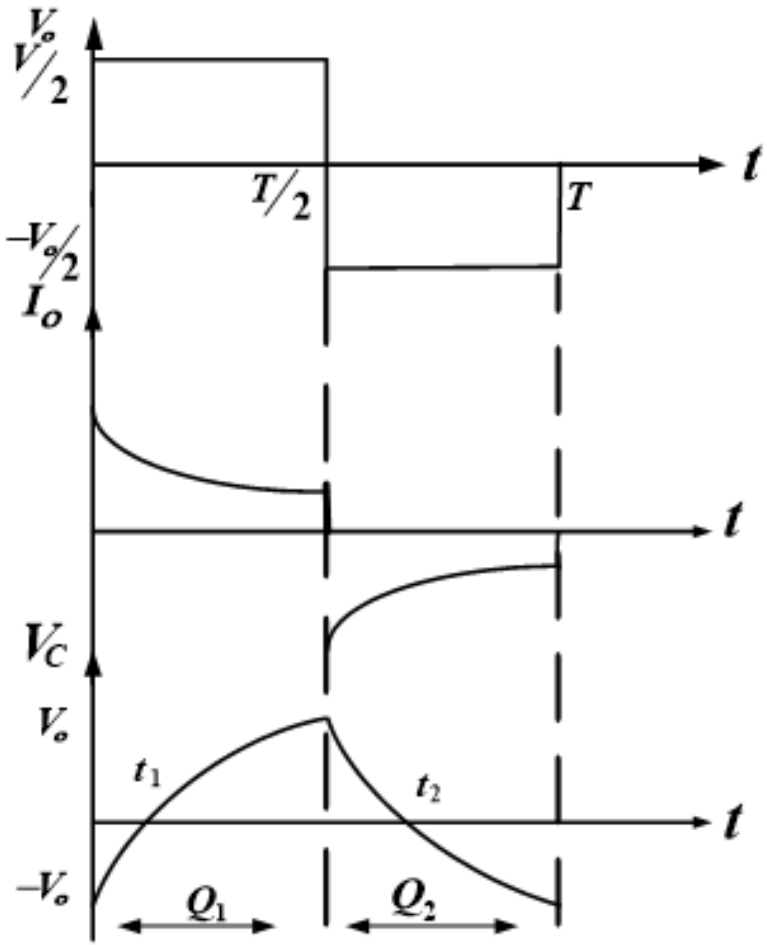
$$I_o = i_o\left(\frac{T}{2}\right) = -\frac{V_S}{2R}\frac{\left[1 - e^{-\frac{R}{2L}T}\right]}{\left[1 + e^{-\frac{R}{2L}T}\right]}$$

Substitute the value of I_o in $i_o(t)$ equation

$$i_o(t') = -\frac{V_S}{2R}\left[1 - \frac{2\,e^{-\frac{R}{L}t'}}{1 + e^{-\frac{R}{2L}T}}\right]$$

Single-Phase half-bridge RC Load Inverter





$$i_o(t) = C\frac{dV_c(t)}{dt}$$

$$\frac{V_s}{2} = R.C\frac{dV_c(t)}{dt} + V_c(t)$$

$$V_c(t) = \frac{V_s}{2}\left[1 - e^{-t/R.C}\right] - V_o e^{-t/R.C}$$

At $t = \frac{T}{2}$ the voltage is $V_C\left(\frac{T}{2}\right) = V_o$

$$V_o = \frac{V_S}{2}\frac{\left[1 - e^{-\frac{T}{2R.C}}\right]}{\left[1 + e^{-\frac{T}{2R.C}}\right]}$$

$$V_c(t) = \frac{V_s}{2}\left[1 - \frac{2}{1 + e^{-\frac{T}{2R.C}}} e^{-t/R.C}\right]$$

$$i_o(t) = C\frac{dV_c(t)}{dt} = \frac{V_s}{R}\left[\frac{e^{-t/R.C}}{1 + e^{-\frac{T}{2R.C}}}\right]$$

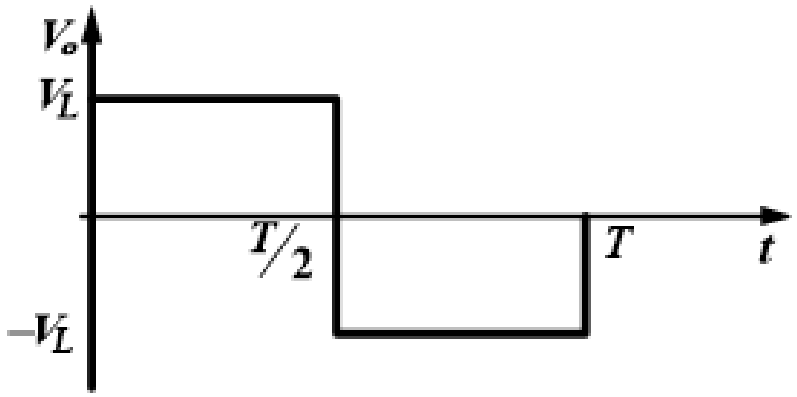
$$T/2 < t < T$$

$$t' = t - \frac{T}{2}$$

$$V_C(t') = -\frac{V_S}{2}\left[1 - \frac{2e^{-t'/R.C}}{1 + e^{\frac{T}{2R.C}}}\right]$$

$$i_o(t') = -\frac{V_S}{R}\left[\frac{e^{-t'/R.C}}{1 + e^{\frac{T}{2R.C}}}\right]$$

Fourier Analysis



$$v_o(t) = \sum_{n=1}^{\infty} a_n Cos(n\omega t) + \sum_{n=1}^{\infty} b_n Sin(n\omega t)$$

$$a_n = \frac{2}{T}\int_0^T v_o(t) Cos(n\omega t)\, d\omega t = \frac{1}{\pi}\int_0^{2\pi} v_o(t) Cos(n\omega t)\, d\omega t$$

$$b_n = \frac{2}{T}\int_0^T v_o(t) Sin(n\omega t)\, d\omega t = \frac{1}{\pi}\int_0^{2\pi} v_o(t) Sin(n\omega t)\, d\omega t$$

Due to the square wave symmetry along the ***x-axis*** ,both a_0 and a_n are ***zero***, and b_n is obtained as:

$$b_n = \frac{1}{\pi}\int_0^{2\pi} v_o(t)Sin(n\omega t)d\omega t = \frac{1}{\pi}\int_0^{\pi} v_L Sin(n\omega t)d\omega t$$

$$+\frac{1}{\pi}\int_{\pi}^{2\pi}(-v_L)Sin(n\omega t)d\omega t$$

$$b_n = \frac{4V_L}{n\pi} \qquad for \quad n = 1,3,5,.....$$

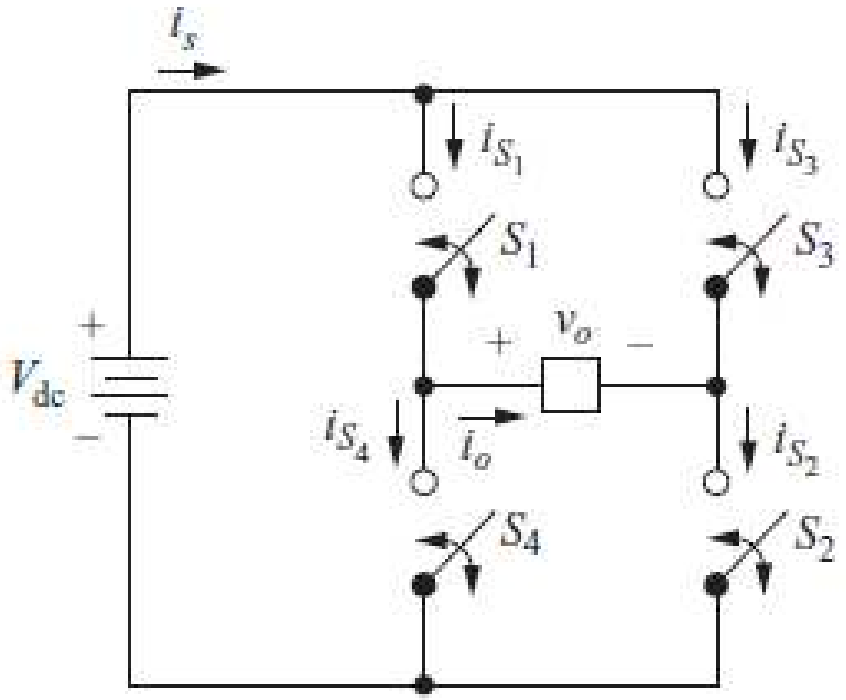
$$b_n = 0 \qquad for \quad n = 2,4,6,.....$$

$$v_o(t) = \sum_{n=1,3,5}^{\infty} \frac{4V_L}{n\pi} Sin(n\omega t)$$

$$= \frac{4V_L}{\pi}\left[Sin\ \omega t + \frac{1}{3}Sin(3\omega t) + \frac{1}{5}Sin(5\omega t) + ...\right]$$

The full-bridge inverter

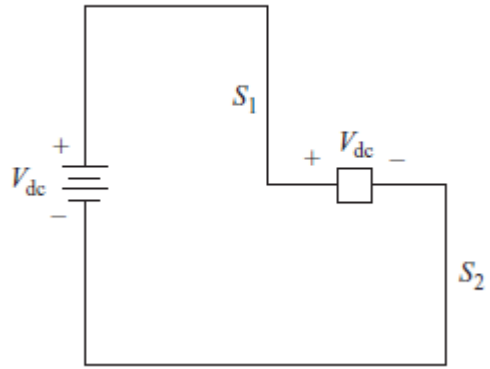
- The basic circuit of the full-bridge inverter:

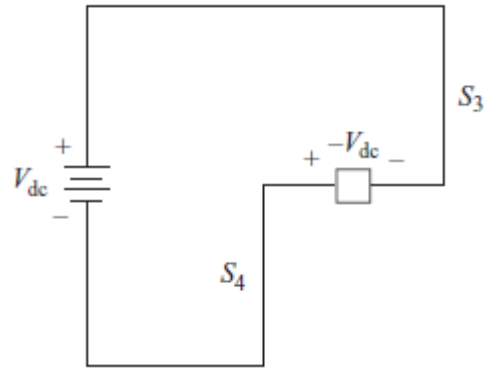


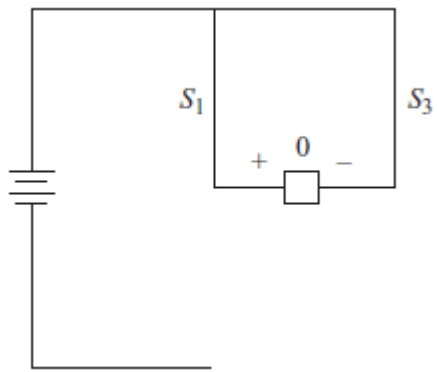
Switching pattern for the full-wave bridge inverter

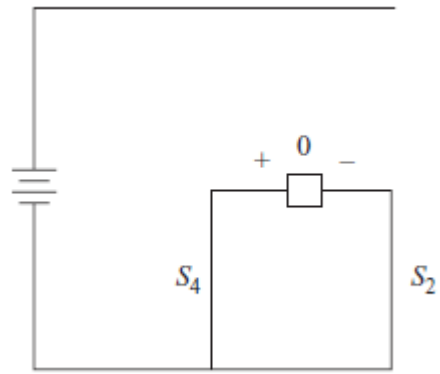
- The output voltage v_o can be $+V_{dc}$, $-V_{dc}$, or zero, depending on which switches are closed. As shown in the following table:

Switches Closed	Output Voltage v_o
S1 and S2	$+V_{dc}$
S3 and S4	$-V_{dc}$
S1 and S3	0
S2 and S4	0


S_1
V_{dc}
S_2


S_3
V_{dc}
$-V_{dc}$
S_4


S_1
S_3
0


0
S_4
S_2

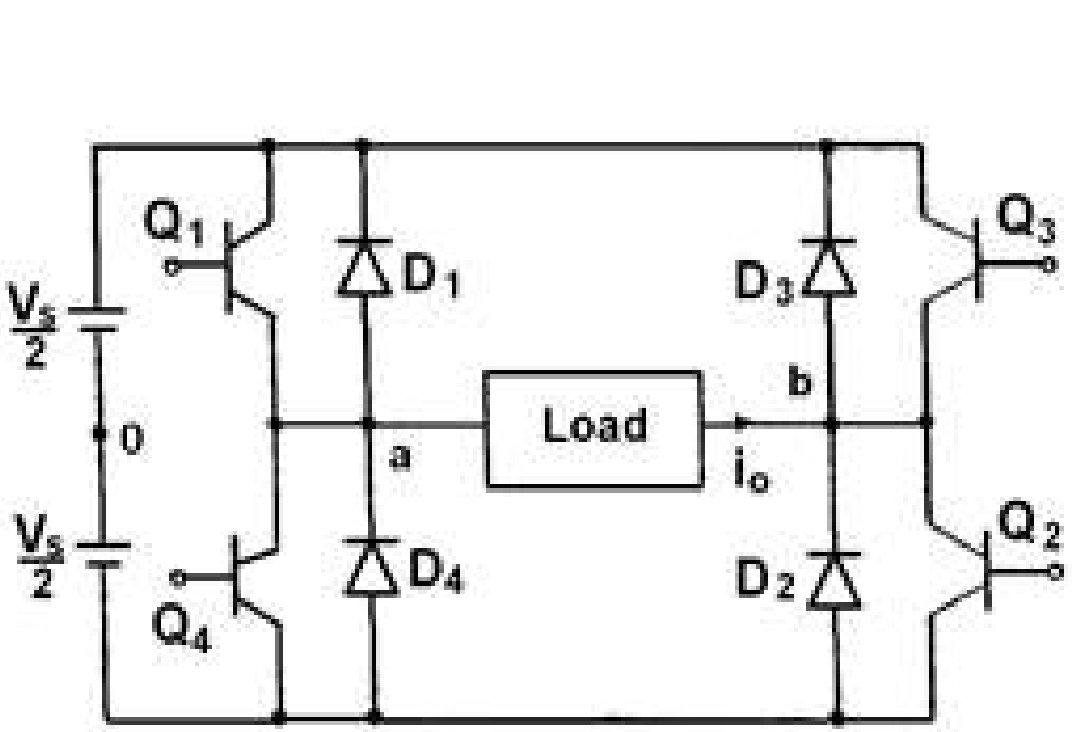
Note that S_1 and S_4, also S_2 and S_3 should not be closed at the same time. Otherwise, a short circuit would exist across the dc source.

Real switches do not turn on or off instantaneously. Therefore, switching transition times must be accommodated in the control of the switches.

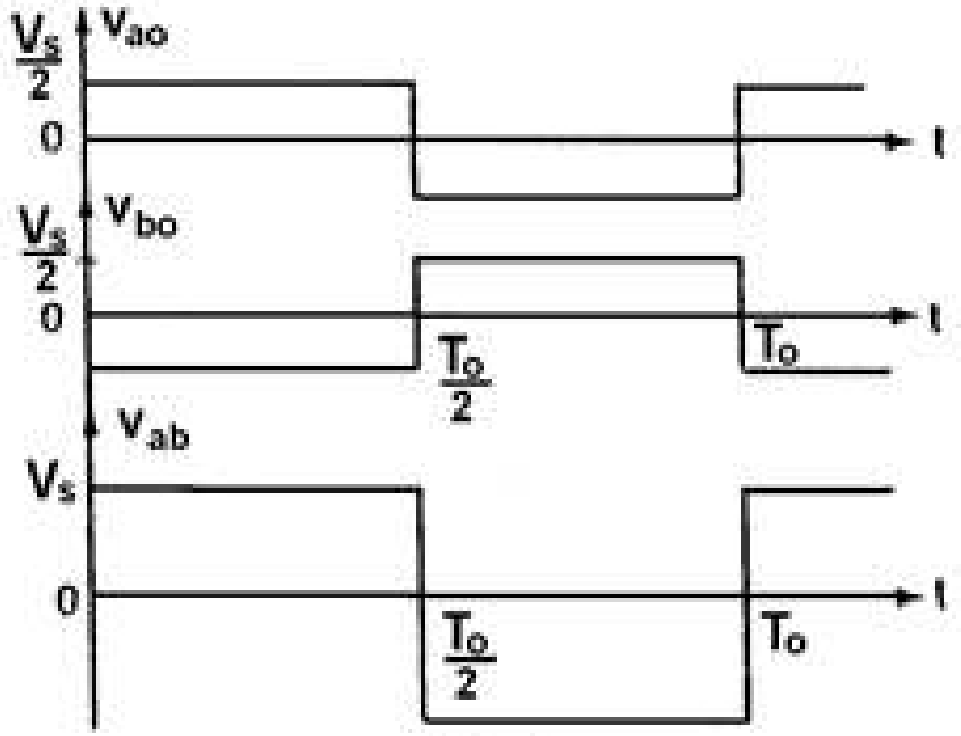
Overlap of switch "on" times will result in a short circuit, sometimes called a *shoot-through* fault, across the dc voltage source. The time allowed for switching is called *blanking* time.

Single-Phase Full- Bridge Voltage Source Inverters with R Load (The square-wave inverter)

- The simplest switching scheme for the full-bridge converter produces a square wave output voltage. The switches connect the load to V_{dc} when S_1 and S_2 are closed or to $-V_{dc}$ when S_3 and S_4 are closed. The periodic switching of the load voltage between V_{dc} and $-V_{dc}$ produces a square wave voltage across the load. Although this alternating output is nonsinusoidal, it may be an adequate ac waveform for some applications.
- The current waveform in the load depends on the load components. For the resistive load, the current waveform matches the shape of the output voltage.



Circuit

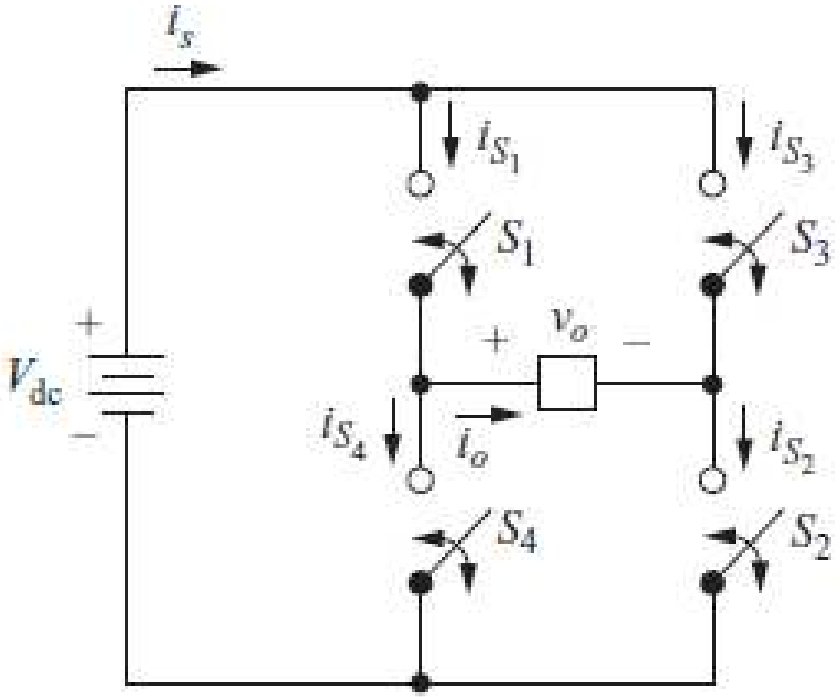


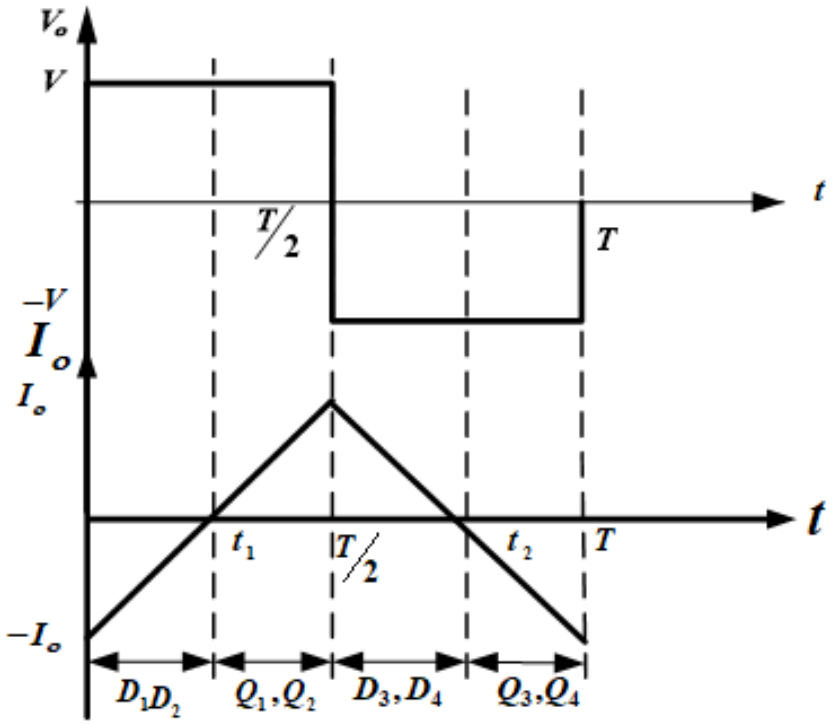
Waveforms

Single-Phase Full- Bridge Voltage Source Inverters with R L Load

An inductive load will have a current that has more of a sinusoidal quality than the voltage because of the filtering property of the inductance. An inductive load presents some considerations in designing the switches in the full-bridge circuit because the switch currents must be bidirectional.

For a series RL load and a square wave output voltage, assume switches $S_1 S1$ and S_2 in the Figure below close at $t = 0$. The voltage across the load is V_{dc}, and current begins to increase in the load and in S_1 and S_2. The current is expressed as the sum of the forced and natural responses:





$$i_o(t) = i_f(t) + i_n(t)$$

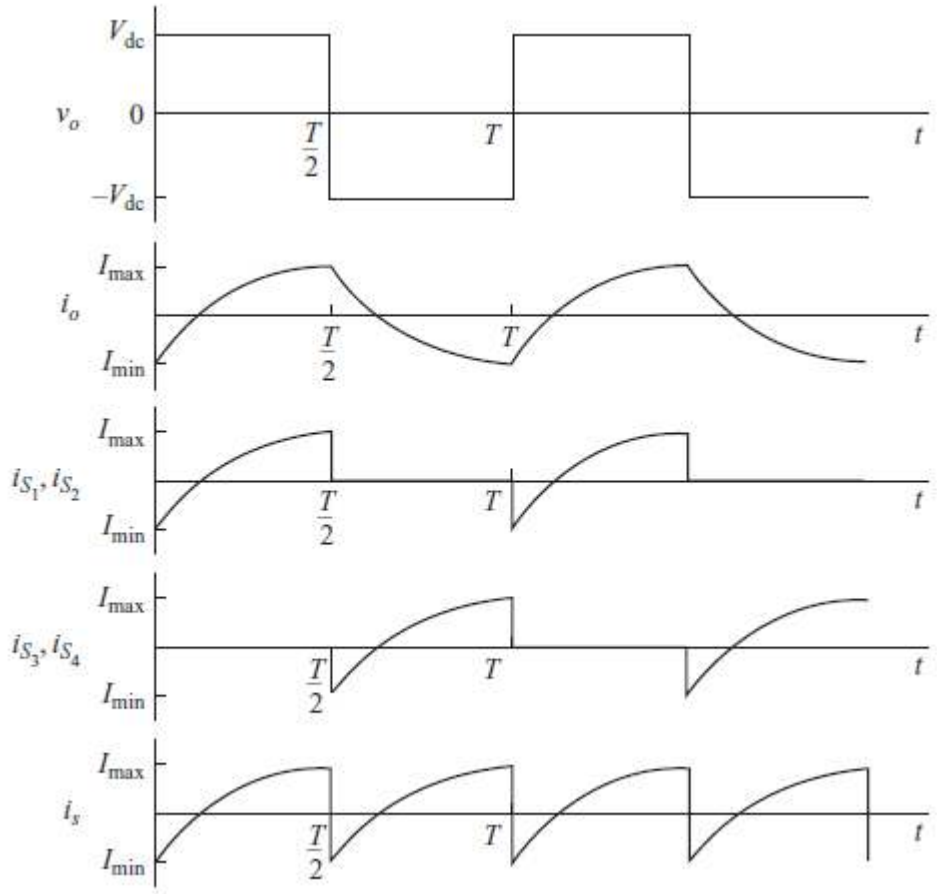
$$= \frac{V_{dc}}{R} + Ae^{-t/\tau} \qquad \text{for } 0 \leq t \leq T/2$$

where A is a constant evaluated from the initial condition and $\tau = \frac{L}{R}$. At $t = T/2$, S_1S1 and S_2 open, and S_3 and S_4 close. The voltage across the RL load becomes $-V_{dc}$, and the current has the form:

$$i_o(t) = \frac{-V_{dc}}{R} + Be^{-(t-T/2)/\tau} \qquad \text{for } T/2 \leq t \leq T$$

where the constant B is evaluated from the initial condition.

When the circuit is first energized and the initial inductor current is zero, a transient occurs before the load current reaches a steady-state condition. At steady state, i_o is periodic and symmetric about zero, as illustrated in the following Figure:



Let the initial condition for the current described in the Equation:

$$i_o(t) = i_f(t) + i_n(t)$$

$$= \frac{V_{dc}}{R} + Ae^{-t/\tau} \qquad \text{for } 0 \leq t \leq T/2$$

be I_{min}.

and let the initial condition for the current described in the Equation:

$$i_o(t) = \frac{-V_{dc}}{R} + Be^{-(t-T/2)/\tau} \qquad \text{for } T/2 \leq t \leq T$$

be I_{max}.

Evaluating the first Equation at $t = 0$, gives:

$$i_o(0) = \frac{V_{\text{dc}}}{R} + Ae^0 = I_{\text{min}}$$

$$A = I_{\text{min}} - \frac{V_{\text{dc}}}{R}$$

Evaluating the 2nd Equation at $t = \frac{T}{2}$, gives:

$$i_o(T/2) = \frac{-V_{\text{dc}}}{R} + Be^0 = I_{\text{max}}$$

$$B = I_{\text{max}} + \frac{V_{\text{dc}}}{R}$$

In the steady state, the current waveforms are described by these Equations then become:

$$i_o(t) = \begin{cases} \dfrac{V_{dc}}{R} + \left(I_{min} - \dfrac{V_{dc}}{R}\right)e^{-t/\tau} & \text{for } 0 < t < \dfrac{T}{2} \\ \dfrac{-V_{dc}}{R} + \left(I_{max} + \dfrac{V_{dc}}{R}\right) - e^{(t-T/2)/\tau} & \text{for } \dfrac{T}{2} < t < T \end{cases}$$

An expression is obtained for I_{max} by evaluating the Equations at $t = \frac{T}{2}$

$$i(T/2) = I_{\max} = \frac{V_{\mathrm{dc}}}{R} + \left(I_{\min} - \frac{V_{\mathrm{dc}}}{R} \right) e^{-(T/2\tau)}$$

But $I_{max} = -I_{min}$

That gives:

$$I_{\max} = -I_{\min} = \frac{V_{\mathrm{dc}}}{R} \left(\frac{1 - e^{-T/2\tau}}{1 + e^{-T/2\tau}} \right)$$

Power absorbed by the load can be determined from $I_{rms}^2 R$. The integration may be simplified by taking advantage of the symmetry of the waveform. Since the square each of the current half-periods is identical, only the first half-period needs to be evaluated:

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,d(t)} = \sqrt{\frac{2}{T}\int_0^{T/2} \left[\frac{V_{\text{dc}}}{R} + \left(I_{\text{min}} - \frac{V_{\text{dc}}}{R}\right)e^{-t/\tau}\right]^2 dt}$$

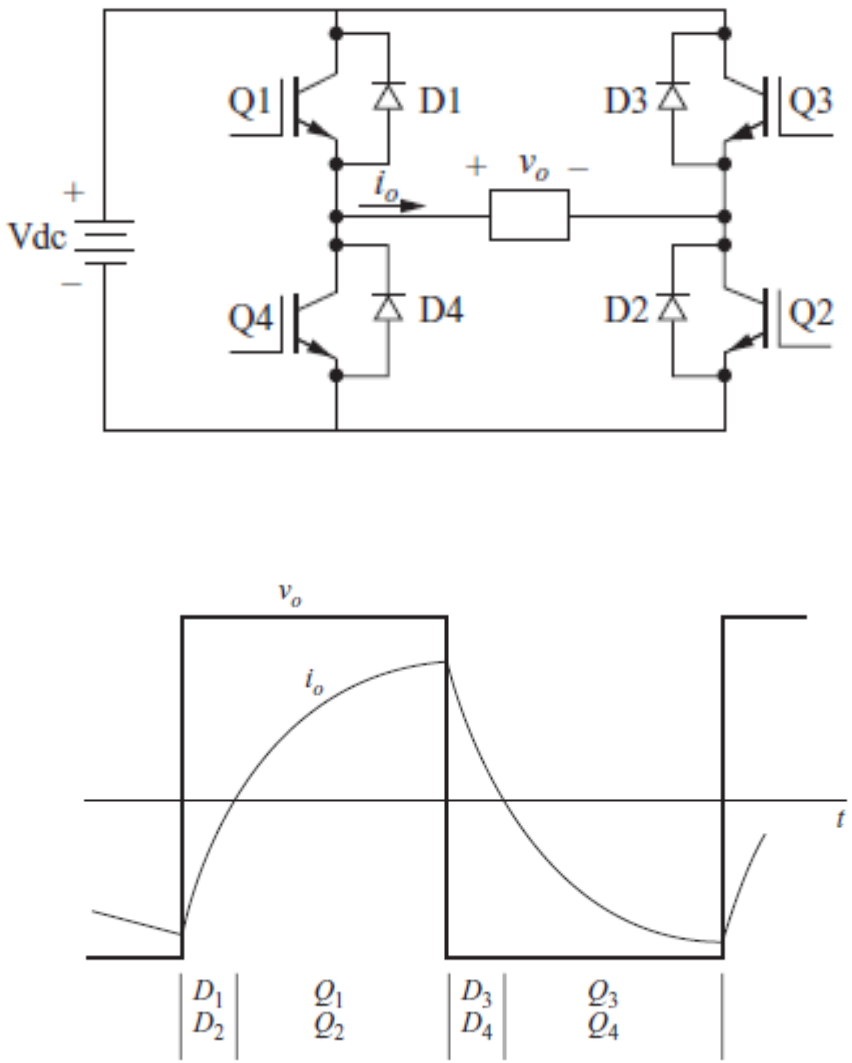
If the switches are ideal, the power supplied by the source must be the same as absorbed by the load.
Power from a dc source is determined from:

$$P_{\text{dc}} = V_{\text{dc}} I_s$$

The switch currents in the bridge circuit show that the switches in the full-bridge circuit must be capable of carrying both positive and negative currents for *RL* loads.

However, real electronic devices may conduct current in one direction only. This problem is solved by placing feedback diodes in parallel (anitparallel) with each switch. During the time interval when the current in the switch must be negative, the feedback diode carries the current. The diodes are reverse-biased when current is positive in the switch. The following Figure shows the full-bridge inverter with switches implemented as insulated gate bipolar transistors (IGBTs) with feedback diodes. Transistor and diode currents for a square wave voltage and an *RL* load are indicated in the Figure. Power semiconductor modules often include feedback diodes with the switches.

When IGBTs Q_1 and Q_2 are turned off in the Figure, the load current must be continuous and will transfer to diodes D_3 and D_4, making the output voltage $-V_{dc}$, effectively turning on the switch paths 3 and 4 before Q_3 and Q_4 are turned on.
IGBTs Q_3 and Q_4 must be turned on before the load current decays to zero.



Fourier series analysis

- The Fourier series method is often the most practical way to analyze load current and to compute power absorbed in a load, especially when the load is more complex than a simple resistive or *RL* load. Auseful approach for inverter analysis is to express the output voltage and load current in terms of a Fourier series. With no dc component in the output:

$$v_o(t) = \sum_{n=1}^{\infty} V_n \sin(n\omega_0 t + \theta_n)$$

$$i_o(t) = \sum_{n=1}^{\infty} I_n \sin(n\omega_0 t + \phi_n)$$

Power absorbed by a load with a series resistance is determined from $I_{rms}^2\ R$, where the rms current can be determined from the rms currents at each of the components in the Fourier series by:

$$I_{\text{rms}} = \sqrt{\sum_{n=1}^{\infty} I_{n,\text{rms}}^2} = \sqrt{\sum_{n=1}^{\infty} \left(\frac{I_n}{\sqrt{2}}\right)^2}$$

where

$$I_n = \frac{V_n}{Z_n}$$

and Z_n is the load impedance at harmonic n.

Equivalently, the power absorbed in the load resistor can be determined for each frequency in the Fourier series. Total power can be determined from:

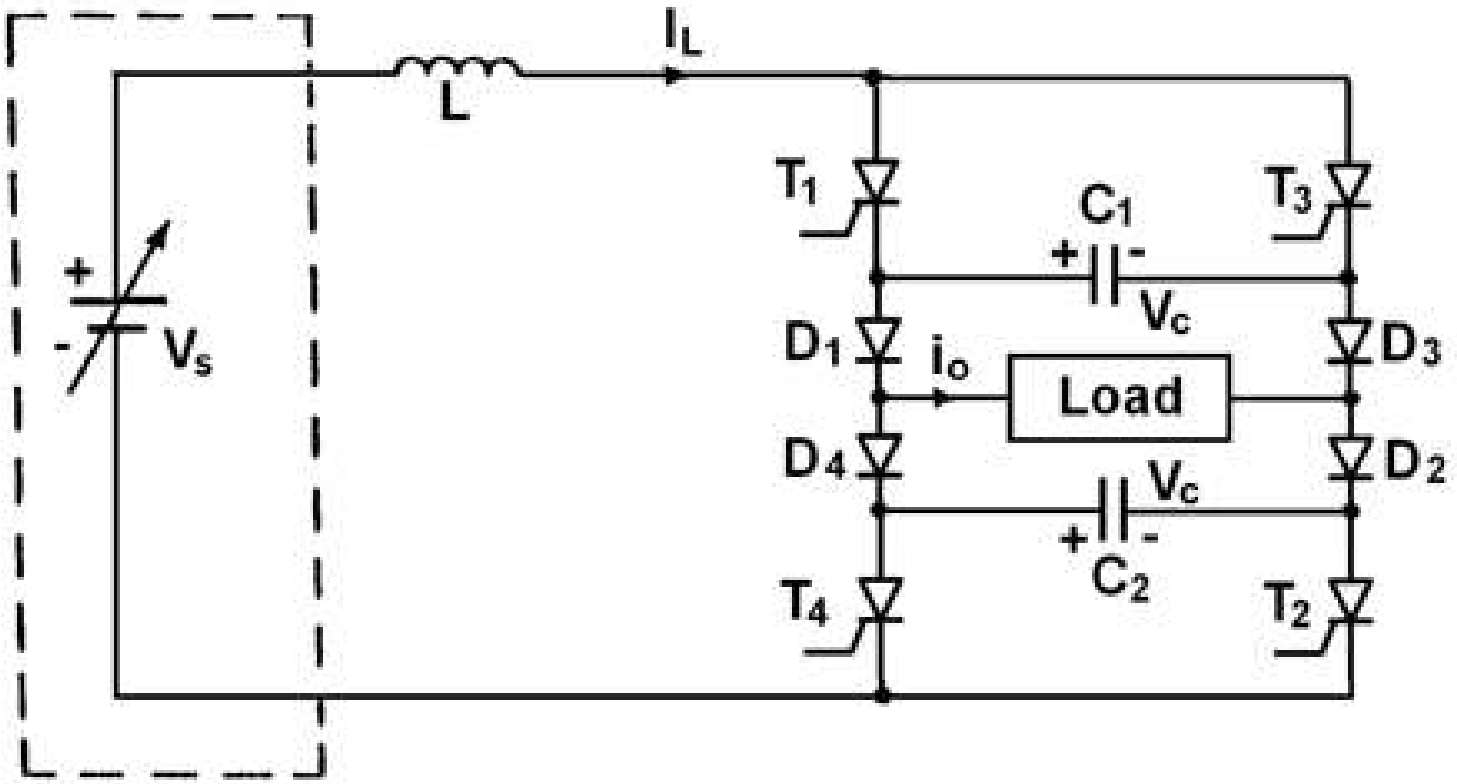
$$P = \sum_{n=1}^{\infty} P_n = \sum_{n=1}^{\infty} I_{n,\text{rms}}^2 R$$

where $I_{n,rms}$ is $\frac{I_n}{\sqrt{2}}$

In the case of the square wave, the Fourier series contains the odd harmonics and can be represented as:

$$v_o(t) = \sum_{n\,\text{odd}} \frac{4V_{\text{dc}}}{n\,\pi} \sin n\omega_0 t$$

Single-Phase Current Source Bridge



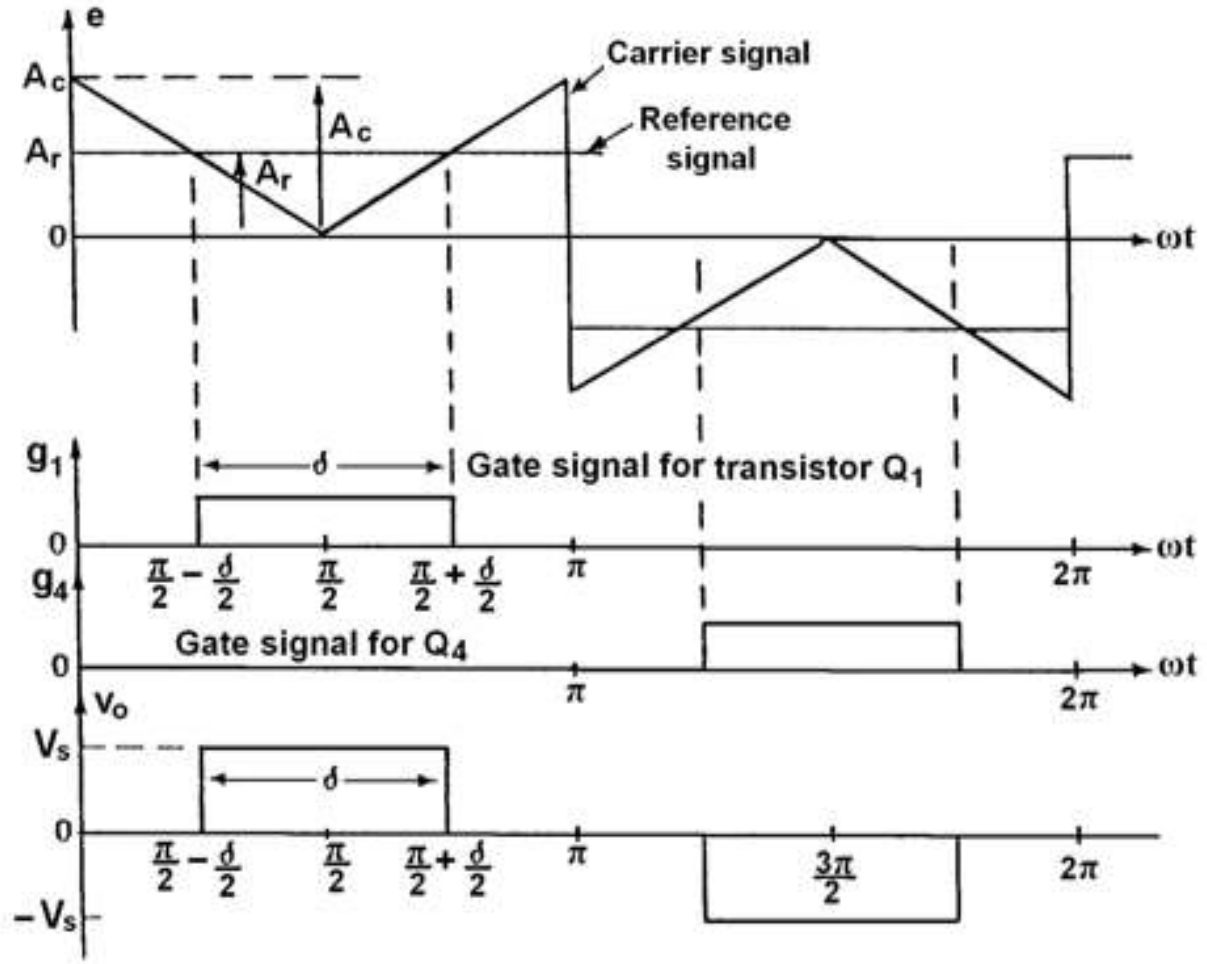
Voltage Control of Single- Phase Inverters

1- Input voltage control:

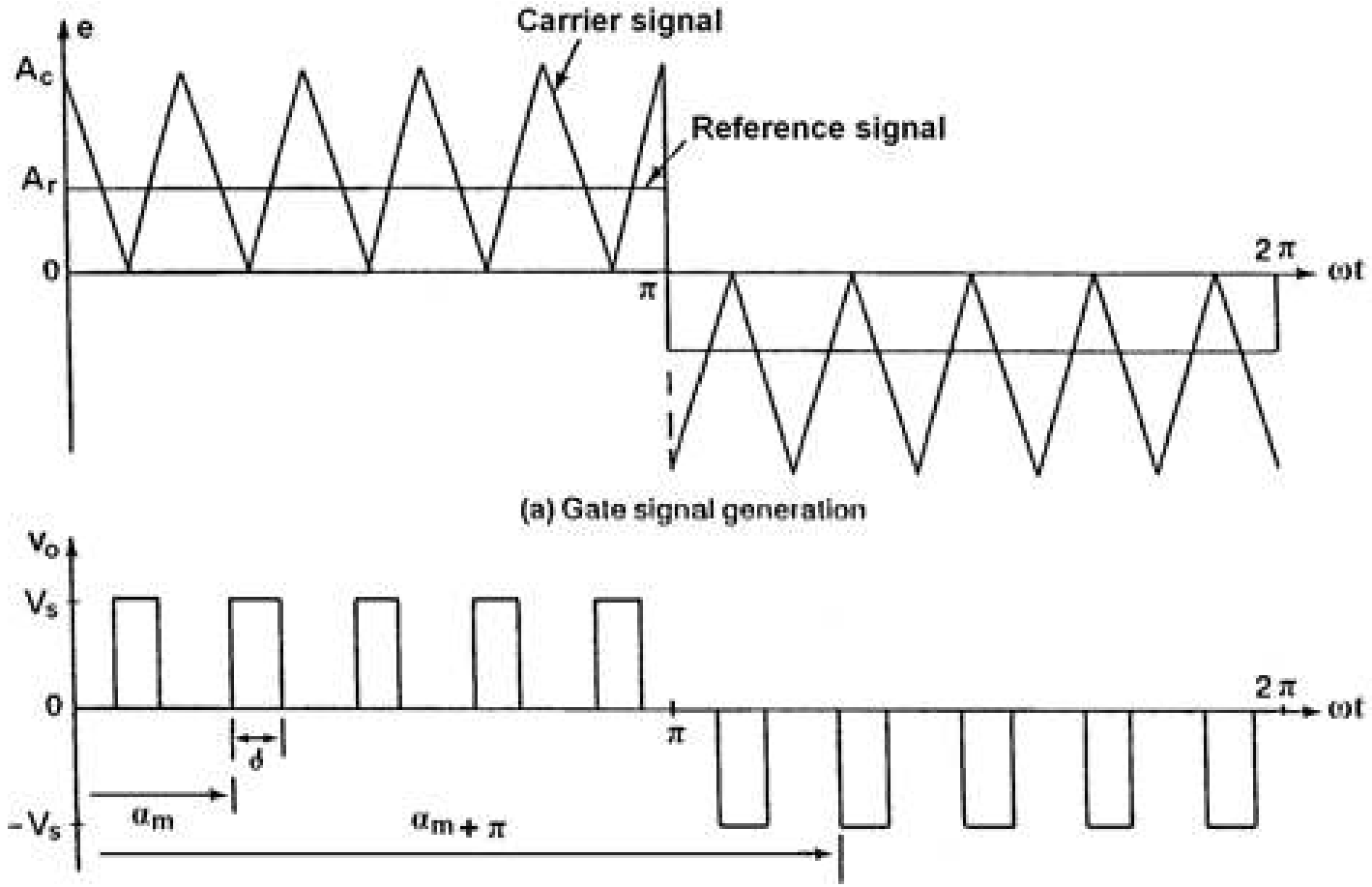
- Variable voltage source.
- Controlled rectifier circuit.

2- Output voltage control.

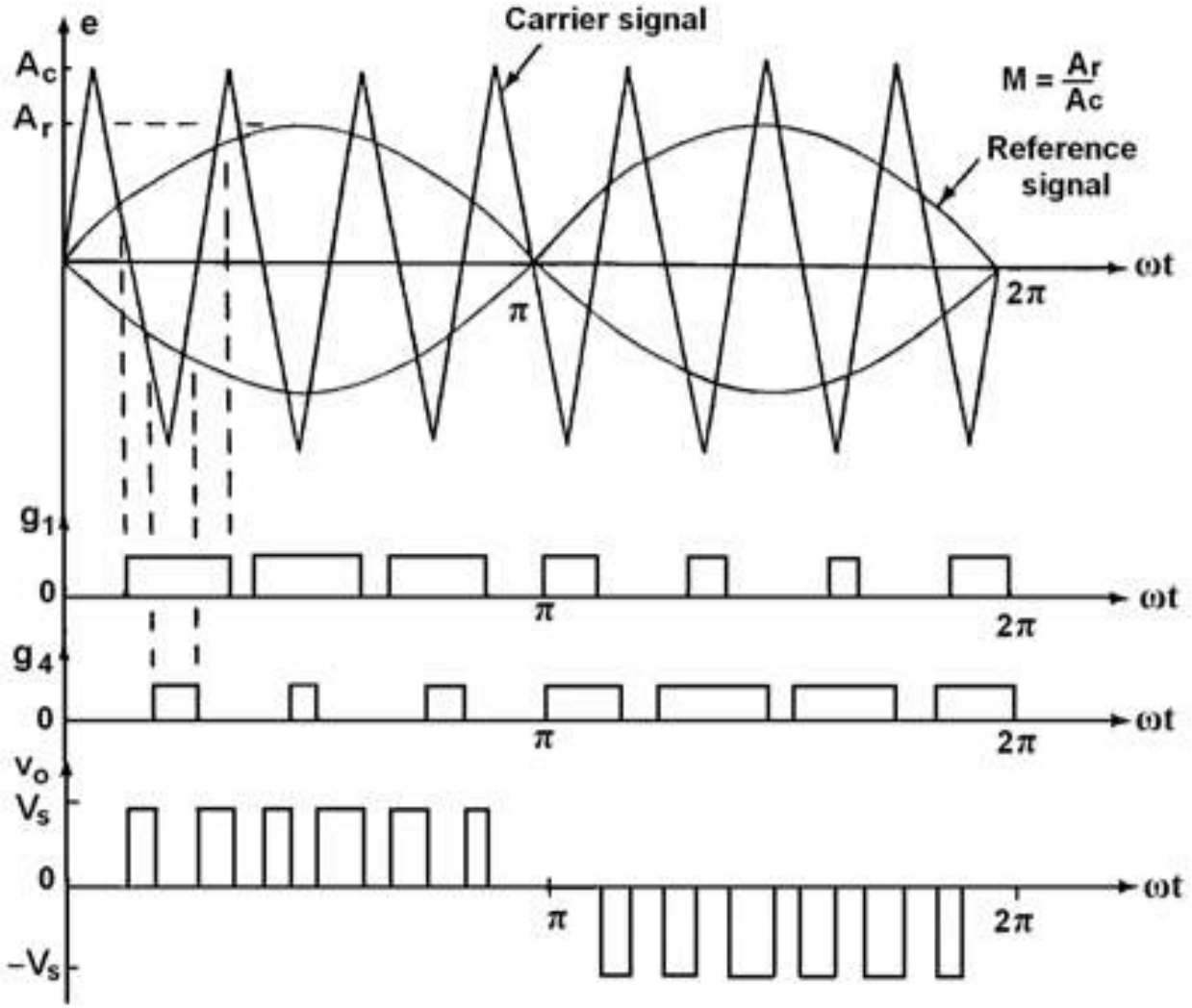
- **Single Pulse-Width Modulation**



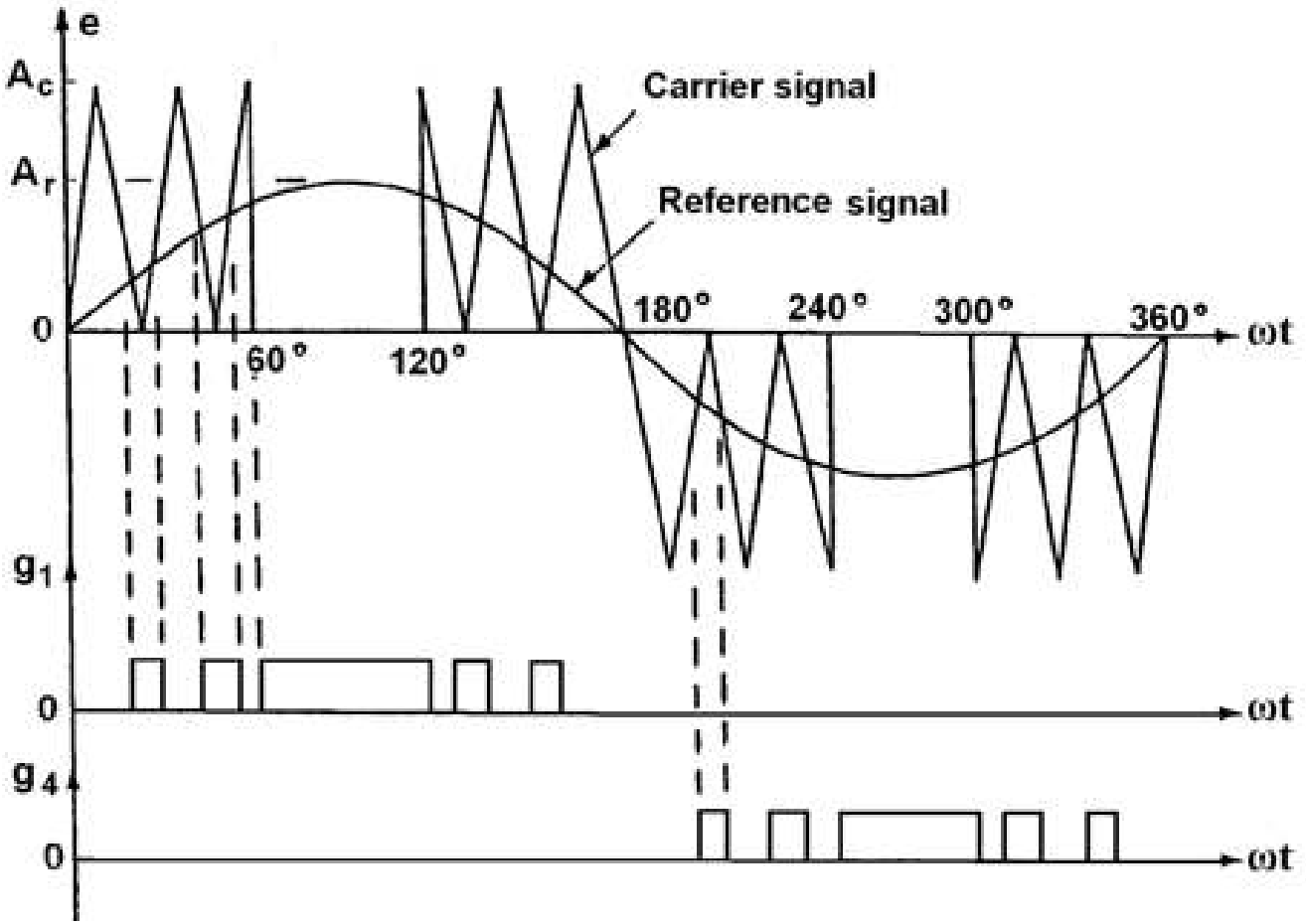
- **Multiple-Pulse-Width Modulation**



Sinusoidal Pulse-Width-Modulation (SPWM)



- **Modified Sinusoidal-Pulse-Width Modulation (MSPUM)**



- Phase-Displacement Control

